

# PERFORM BASIC FIRE FIGHTING

## FACILITATOR/LEARNER GUIDE

Unit Standard No: 12184  
Unit Standard Credits: 4  
NQF Level: 2

SKILLS PROGRAM 1  
BOOK 8



| PROGRAM GUIDE                        |                                                             | 5-6          |
|--------------------------------------|-------------------------------------------------------------|--------------|
| 1.                                   | Introduction                                                | 5            |
| 2.                                   | Purpose of this learning program                            | 5            |
| 3.                                   | Standards and qualifications                                | 5            |
| 4.                                   | Assessment                                                  | 5            |
| 5.                                   | Security program matrix                                     | 6            |
| <b>Study unit 1: Types of Fires</b>  |                                                             | <b>7-13</b>  |
| 1.                                   | Definition of a fire                                        | 8            |
| 2.                                   | Elements of a fire                                          | 8            |
| 3.                                   | Classification of fire                                      | 9            |
| 4.                                   | Conditions of a fire                                        | 10           |
| 5.                                   | Common sources of heat                                      | 11           |
| 6.                                   | Transmission ways in which a fire spread                    | 12           |
| <b>Study unit 2: Fire Prevention</b> |                                                             | <b>14-19</b> |
| 1.                                   | Goals of fire prevention                                    | 15           |
| 2.                                   | Fire hazards                                                | 15           |
| 3.                                   | Preventative measures                                       | 17           |
| <b>Study unit 3: Housekeeping</b>    |                                                             | <b>20-23</b> |
| 1.                                   | Why is good housekeeping important?                         | 21           |
| 2.                                   | General housekeeping guideline                              | 21           |
| 3.                                   | Do's and Don'ts                                             | 22           |
| <b>Study unit 4: Fire Fighting</b>   |                                                             | <b>24-31</b> |
| 1.                                   | Know the use of different types of fire extinguishers       | 25           |
| 2.                                   | How to operate a fire extinguisher                          | 30           |
| 3.                                   | Fire hoses                                                  | 31           |
| <b>Study unit 5: Fighting Fire</b>   |                                                             | <b>32-37</b> |
| 1.                                   | Make the correct decision in the process of fighting a fire | 33           |
| 2.                                   | Fighting a fire                                             | 33           |
| 3.                                   | Fighting the fire                                           | 34           |
| 4.                                   | Procedure after the fire                                    | 35           |
| 5.                                   | Handing over to the emergency services                      | 35           |
| 6.                                   | ROAST                                                       | 36           |
| 7.                                   | Methods of extinguishment                                   | 36           |
| <b>Unit Standard</b>                 |                                                             | <b>38-51</b> |
| <b>Learner Workbook Section</b>      |                                                             | <b>52-58</b> |

## REVIEW AND ALTERATION OF TRAINING MATERIAL

What follows is a brief explanation of the process that was followed in the alteration of this training material. A working group was formed after inviting interested parties to attend a meeting at SASSETA.

The working group was mandated to review the material and thereafter make the necessary changes so as to provide the industry with a more user friendly set of materials which better reflects the needs of the security industry.

It must be understood that this is the first review and by no means the final review. The working group was placed under enormous pressure to get a workable set of materials into the hands of the accredited security industry training providers as quickly as possible. We therefore have no doubt that even though the materials have been vastly improved upon, there are still areas that may require change. This we plan to do in the next renewal phase after we receive feedback from training providers who have used the material for approximately a year.

Our review process focused on the following:

- Removal of unnecessary information/duplication from the learning material.
- Ensure alignment with the unit standards.
- Re-draft all formative and summative assessments.
- Correct inappropriate use of language.

### TASK TEAM

The task team that completed the work on this training material deserves a very special "thank you", considering that all their time and efforts were provided free of charge. Nobody was paid for any of the work done on behalf of the task team. SASSETA provided funding for the expenses incurred in printing, typesetting, lunch and refreshments.

The task team members are as follows:

|                              |                                        |
|------------------------------|----------------------------------------|
| Andre Pretorius              | International Firearm Training Academy |
| Andre Wilken                 | SSN                                    |
| Dave Dodge                   | ESKOM                                  |
| Eddie Du Plooy/ Elvis Masera | S&V Services                           |
| Jean Du Plessis              | Lytleton Firearm Training Centre       |
| Leon van Rooyen              | N/D                                    |
| Lionel Arries                | SASSETA (observer/adviser)             |
| Marion Colley                | Pecoo Security and Training            |

A very sincere thank you to all of these individuals and the companies they work for, who allowed them to participate during business hours. This could not have been done without your commitment.

Sincerely

Andre Pretorius  
Task Team Chairman

5. Security program matrix

| SKILLS PROGRAM 1: SASSETA E |        |                                                                   |                      |
|-----------------------------|--------|-------------------------------------------------------------------|----------------------|
| 1                           | 246694 | Explain the requirements for becoming a security service provider | Level 3<br>4 Credits |
| 2                           | 244184 | Apply legal aspects in a security environment                     | Level 3<br>8 Credits |
| 3                           | 244182 | Give evidence in court                                            | Level 3<br>4 Credits |
| 4                           | 244176 | Use security equipment                                            | Level 2<br>2 Credits |
| 5                           | 244181 | Perform hand over and take over responsibilities                  | Level 3<br>2 Credits |
| 6                           | 244177 | Conduct a security patrol in area of responsibility               | Level 3<br>7 Credits |
| 7                           | 244179 | Handle complaints and problems                                    | Level 3<br>6 Credits |
| 8                           | 12484  | Perform basic fire fighting                                       | Level 2<br>4 Credits |
| 9                           | 116534 | Carry out basic first aid treatment in the workplace              | Level 3<br>2 Credits |

Patrol Security officer/Grade E

| SKILLS PROGRAM 2: SASSETA D |       |                                                                             |                       |
|-----------------------------|-------|-----------------------------------------------------------------------------|-----------------------|
| 1                           | 24418 | Conduct access and egress control                                           | Level 4<br>7 Credits  |
| 2                           | 24282 | Conduct evacuations and emergency drills                                    | Level 4<br>4 Credits  |
| 3                           | 11505 | Identify, handle and defuse security related conflict                       | Level 4<br>12 Credits |
| 4                           | 11770 | Demonstrate knowledge of the Firearms Control Act 2000 (Act No. 60 of 2000) | Level 3<br>3 Credits  |
| 5                           | 11392 | Apply basic business ethics in a work environment                           | Level 2<br>2 Credits  |
| 6                           | 11946 | Write/present/sign texts for a range of communicative contexts              | Level 3<br>5 Credits  |
| 7                           | 11497 | Operate a computer workstation in a business environment                    | Level 3<br>2 Credits  |

Access control officer/Grade D

| SKILLS PROGRAM 3: SASSETA C |        |                                                                                         |                       |
|-----------------------------|--------|-----------------------------------------------------------------------------------------|-----------------------|
| 1                           | 113809 | Coach a team member in order to enhance individual performance in work environment      | Level 3<br>5 Credits  |
| 2                           | 13812  | Apply knowledge of self and team in order to develop a plan to enhance team performance | Level 3<br>5 Credits  |
| 3                           | 244578 | Describe how to manage reactions arising from a traumatic event                         | Level 3<br>2 Credits  |
| 4                           | 113852 | Apply occupational health, safety and environmental principles                          | Level 3<br>10 Credits |
| 5                           | 13936  | Outline the legal environment of a selected industry                                    | Level 3<br>2 Credits  |
| 6                           | 119472 | Accommodate audience and context needs in oral/sign communication                       | Level 3<br>5 Credits  |
| 7                           | 11508  | Write security reports and take statements                                              | Level 4<br>10 Credits |

Asset & Reaction officer/Grade C

Introduction

This learning program is part of a complete qualification. For more details concerning the complete qualification please contact your service provider or alternatively the office of the SASSETA.

2. Purpose of this learning program

The skills, values and knowledge reflected in this unit standard are required by people in the field of manufacturing and engineering. The learning outcomes in this unit standard also contribute to the exit level outcomes required for various manufacturing and engineering qualifications.

Qualifying learners can select and use the appropriate fire fighting equipment to extinguish or control fires in the workplace.

2.1 Target group

This program is compiled for the following target group:

- Security members.
- South African Defense Force members.
- South African Police Force members.
- Correctional Services.
- Individuals who wish to complete the NQF level 3 National Certificate in security practices.

3. Standards and qualification:

Unit standards are the 'building blocks' of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF).

Unit standards comprises of outcomes. An outcome is a statement that describes the required competency that must be demonstrated by the learner on successful completion of a training intervention.

4. Assessments

The assessment criteria in this unit standard describes the evidence that is needed that will show that you have demonstrated the outcome correctly.

Kindly refer to the unit standard attached hereto for the assessment criteria listed under each Specific Outcome in order for you to see what you will be assessed against.

You will be required to complete 2 written exams. The first is a formative assessment (open book exam) and the second is a summative assessment (closed book exam). The purpose of the formative assessment is to prepare you for the summative assessment.

The learner guide will remain the property of the learner once the LEARNING PROGRAM has been completed.

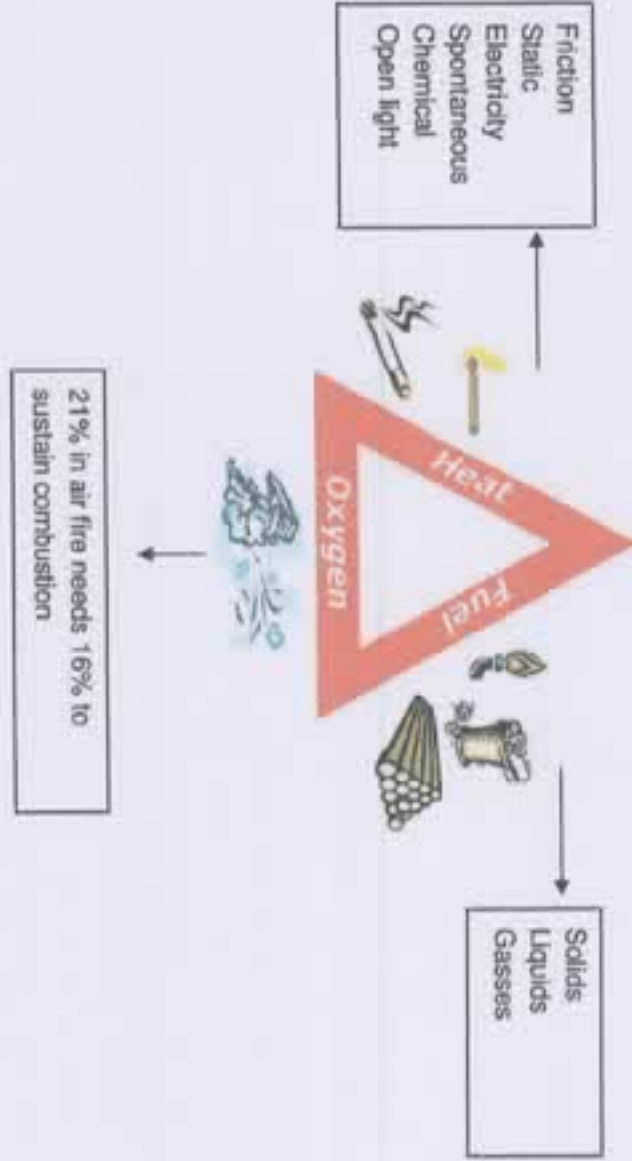
1. Definition of a fire

The state of combustion in which **flammable materials** burns, producing **HEAT, FLAMES** and **SMOKE**.

2. Elements of a fire

For fires to exist, the following four elements must be present at the same time:

- Enough **OXYGEN** to sustain combustion,
- Enough **HEAT** to raise the material to its ignition temperature,
- Some sort of **FUEL** or combustible material, and
- The chemical reaction that is fire.



STUDY UNIT 1

TYPES OF FIRES

LEARNING OUTCOMES

On completion of this study unit the student will....



- Define a fire.
- Identify the elements of a fire.
- Demonstrate knowledge of the different types of fire.
- Identify the various causes of a fire.



- **Class D:** Fires involving combustible metals such as magnesium, sodium, potassium, titanium, and aluminum. Special dry powder extinguishing agents are required for this class of fire, and must be tailored to the specific hazardous metal.

F



- **Class F:** Fires involving commercial cooking appliances with vegetable oils, animal oils, or fats at high temperatures. A wet potassium acetate, low pH-based agent is used for this class of fire.

#### 4. Conditions of a fire (conditions that you can expect)

- Darkness
- Heat
- Smoke



##### **Fire is FAST!**

##### **There is little time!**

In less than 30 seconds a small flame can get completely out of control and turn into a major fire. It only takes minutes for thick black smoke to fill a room. In minutes, a building can be engulfed in flames. You won't have time to grab valuables because fire spreads too quickly and the smoke is too thick. There is only time to escape.

##### **Fire is HOT!**

##### **Heat is more threatening than flames**

A fire's heat alone can kill. Room temperatures in a fire can be 100 degrees at floor level and rise to 600 degrees at eye level. Inhaling this super hot air will scorch your lungs. This heat can melt clothes to your skin. In five minutes a room can get so hot that everything in it ignites at once: this is called flashover.



Demonstrate knowledge of the different types of fire.

### 3. Classification of fire.

Portable fire extinguishers are classified to indicate their ability to handle specific classes and sizes of fires. Labels on extinguishers indicate the class and relative size of fire that they can be expected to handle. Most fires that occur will fall into one or more of the following categories:

A



- **Class A:** Fires involving ordinary combustible materials, such as paper, wood, and textile fibres, where a cooling, blanketing, or wetting extinguishing agent is needed.

B



- **Class B:** Fires involving flammable liquids such as gasoline, thinners, oil-based paints and greases. Extinguishers for this type of fire include carbon dioxide, dry chemical and halogenated agent types.

C



- **Class C:** Fires involving energized electrical equipment, where a non-conducting gaseous clean agent or smothering agent is needed. The most common type of extinguisher for this class is a carbon dioxide extinguisher.



## 6. Transmission ways in which a fire spread

- Conduction.
- Convection.
- Radiation.



### Radiation

Radiation refers to the emission of energy in rays or waves. Heat moves through space as energy waves. It is the type of heat one feels when sitting in front of a fireplace or around a campfire. It travels in straight lines at the speed of light. This is the reason that when facing the fire, only the front is warmed. The backside is not warmed until the person turns around. The earth is heated by the sun through radiation. Sunburns are a "fact of life" when people are exposed to the sun very long. Most of the preheating of fuels ahead of a fire is by radiation of heat from the fire. As the fire front gets closer, the amount of radiant heat increases and heats the combustible materials.



### Fire is DARK!

**Fire isn't bright, it is pitch black**

Fire starts bright, but quickly produces black smoke and complete darkness. You may be blinded, disoriented and unable to find your way around even in your own home you have lived in for years.

**Fire is DEADLY!**  
Smoke and toxic gases kill more people than flames do. Fire uses up the oxygen you need and produces smoke and poisonous gases that kill. Breathing even small amounts of smoke and toxic gases can make you drowsy, disoriented and short of breath. The odourless, colourless fumes can lull you into a deep sleep before the flames reach your door. You may not wake up in time to escape. CO (Carbon Monoxide) is often referred to as the silent killer. It is a colourless, odourless, tasteless, non-irritating, toxic gas.



## 5. Common sources of heat

- Electricity.
- Smoking.
- Friction.
- Cutting and welding.
- Spontaneous ignition.
- Static electricity.
- Chemical reaction.



# STUDY UNIT 2

## FIRE PREVENTION

### LEARNING OUTCOMES

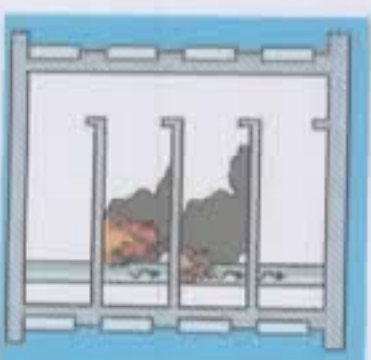
On completion of this study unit the student will....

- Demonstrate knowledge of the goals of preventing a fire.
- Demonstrate skills and knowledge to setup a strategy to prevent a fire.
- List general measures to prevent a fire.



#### Convection

Convection is the transfer of heat by the physical movement of hot masses of air. As air is heated, it expands (as do all objects). As it expands, it becomes lighter than the surrounding air and it rises. (This is why the air near the ceiling of a heated room is warmer than that near the floor.) The cooler air rushes in from the sides. It is heated in turn and it also rises. Soon a convection column is formed above the fire which can be seen by the smoke that is carried aloft in it. This "in-draft" of cooler air from the side helps to supply additional oxygen for the combustion process to continue.



#### Conduction

Conduction is the transfer of heat within the material itself. Most metals are good heat conductors. Wood is a very poor conductor so it transmits heat very slowly. This can be illustrated by the fact that a wooden handle on a hot frying pan remains cool enough to be held by the bare hands. Conduction is not an important factor in the spread of forest fires.



### 2.3. Friction hazards

- Hot bearings.
- Misaligned or broken machine parts.
- Choking or jamming materials.
- Poor adjustment of moving parts.
- Inadequate lubrication.

### 2.4. Process or operation-related hazards

- Cutting and welding operations, which use open flames and produce sparks.
- Molten metal, which can ignite combustibles or fall into cracks and start a fire that might not erupt after the work is done.
- Processes that heat materials to high temperatures.
- Drying operations where materials in dryers can overheat.
- Grinding operations that produce sparks and dust.
- Processes in which flammable vapours are released.



### 2.5. Storage hazards

- Material stacked too high blocking sprinkler heads (need 18-inches clearance from head).
- Flammable or combustible material stored too close to heat sources.
- Flammable materials not stored in special containers and cabinets. Inadequate ventilation in storage areas.
- Material that might react with one another stored together.
- Material stored in damaged containers.
- Material stored in unlabeled containers.
- Containers not tightly sealed.



## Demonstrate knowledge of the goals of preventing a fire.

Fire fighting is a very important job. But if you think about it, fire prevention is even more crucial. Why fight something that you can prevent in the first place, right? We always think about fire protection equipment and the use of fire fighting tools but sometimes we neglect the importance of practicing fire preventive measures to prevent an actual fire accident from happening by identifying fire hazards and prevent the triangle from forming.

### 1. GOALS OF FIRE PREVENTION

- **IDENTIFY** fire hazards
- **TO ELIMINATE** fire hazards
- **TO COMPLY** with by-laws

### 2. FIRE HAZARDS

#### 2.1. Electrical hazards

- Damaged wiring.
- Damaged plugs.
- Damp or wet wires.
- Overloaded motors.
- Broken switches, outlets, or sockets.
- Problems with lighting fixtures.
- Faulty heating elements.
- Overloaded circuits.
- Liquids near computers.
- Computers without surge protectors.

#### 2.2. Housekeeping hazards

- Piles of scrap, waste materials, and trash.
- Sawdust, metal or plastic powder that can form an explosive mixture with air.
- Obstructed aisles.
- Blocked emergency exits.
- Material covering up fire extinguishers, exit signs, and alarms.
- Blocked sprinkler heads.



- **Emergency numbers:** While you should call emergency services if there is a fire, do not call from the location of the fire.

Remember to update your emergency number list regularly.

- **Emergency plan:** Make sure everyone is familiar with the evacuation plan and the location of the emergency doors. Should one exit route be blocked by the fire, you must have another exit to use. Agree on a place to meet to do a "head count" once you evacuate the building.

Always clear emergency exit doors and exit routes from unnecessary objects.

- **Smokers:** Never smoke while going to sleep or when you are sleepy.

The ashes could easily start a fire while you are sleeping or too drowsy to be alert.

- **Heaters:** Any portable heater should be kept at least a meter from anything that can burn (including the wall). Never leave a heater on when you are not in the office.

- **Matches:** It is very important to keep matches out of reach of children. Do not keep matches near a fireplace.

Half of all fire deaths occur at night, so fire hazard checks and special attention to fire prevention should occur before applying your mind on something else.

**Cooking:** Keep the cooking areas clear of combustibles. Make sure to keep all pot handles turned inward so they are not accidentally knocked over. Try to wear short sleeves or fitted clothes so you sleeves don't catch fire.

In case a grease fire occurs, carefully and calmly slide a lid over the pot to smother the flames and turn off the stove.

- **Tumble dryer:** Never leave your tumble dryer on when you are not at home. You should clean the dryer frequently to keep the airway clear.

- **Electrical appliances and wiring**

## 2.6. Smoking hazards

- Ignoring "No Smoking" signs.
- Smoking around flammable or combustible materials.
- Throwing matches and cigarettes or cigars on tables or workbenches.
- Tossing butts on the floor or grass without properly extinguishing them in an ashtray or ash can.
- Tossing lighted butts or matches out windows or doors.
- Smoking in bed.
- Leaving a cigarette/cigar unattended.
- Smoking in areas where there is an accumulation of sawdust, plastic or metal powders that may become explosive.



## 3. PREVENTATIVE MEASURES

Since we know that all the three conditions of **HEAT, FUEL AND OXYGEN** must be present in order for a fire to start, our basic fire prevention strategy must be to **KEEP THEM ISOLATED FROM EACH OTHER**. Heat sources must be kept away from combustible materials and if possible, from oxygen.

### 3.1. General measures to prevent a fire

- **Smoke detector:** It is important to have smoke detectors installed on each floor.

Remember to check the smoke detectors regularly for proper function.

## STUDY UNIT 3

### HOUSEKEEPING

#### LEARNING OUTCOMES

On completion of this study unit the student will....

- Demonstrate the importance of housekeeping.
- Apply good housekeeping guidelines according to company SOP.
- Demonstrate knowledge of storage guidelines.



**NEVER OVERLOAD** an electrical receptacle with extension cords or three-way adapters. If you need to plug a lot of things into a single receptacle, use a power strip.

**NEVER RUN ELECTRICAL CORDS UNDER CARPETS** or rugs, over nails, in traffic areas, or where there is chance of wear.

**DON'T MISUSE EXTENSION CORDS.** Never use a standard household extension cord for a major appliance or a heat-producing appliance; instead use a heavy-duty appliance cord that can carry the load.



When purchasing any **ELECTRICAL DEVICE**, choose only one that carries the **SANS RATING**.

**DON'T PULL ON AN ELECTRICAL CORD** to unplug it. Grasp the plug firmly and pull out straight so you don't bend the prongs.

**REPLACE ANY CORD OR PLUG THAT IS CRACKED** or discoloured or feels hot to the touch when in use. If the original wire or plug is polarized (one prong is wider than the other), or if it has a third grounding prong, replace it with similar equipment.

**CALL AN ELECTRICIAN IF YOUR LIGHTS FLICKER** and your appliances run sluggishly, or if you have a fuse that repeatedly blows or a circuit breaker that trips often. These are signs that the system is overloaded.



**DON'T USE A BULB WITH MORE WATTAGE** than a lamp or fixture is rated to accommodate, or the unit will overheat.

### 3. DO'S

- Keep floors clean and clear of waste.
- Keep workplaces adequately lighted
- Keep light fixtures clean.
- Inspect and clean machinery and tools regularly.
- Clean up spills immediately.



### DON'T'S

- Use aisles and stairways as storage areas.
- Block emergency exits, fire equipment or first aid stations with stored materials.
- Allow materials to build up on floors.
- Store compressed gases near heat sources.



### Introduction

Good housekeeping habits are an important part of a safe workplace.

Effective housekeeping can eliminate some workplace hazards and help get a job done easily and properly.

A good housekeeping program plans and manages the orderly storage and movement of materials from point of entry to exit. It includes a material flow plan to ensure minimal handling.

Employees should be reporting any unusual conditions or hazards as well as obeying posted warning signs.

### 1. Why is good housekeeping important?

- To reduce amounts of flammable and combustible materials.
- To reduce ignition hazards.
- To ensure safe emergency evacuation of occupants.
- To allow for quick emergency response.

### 2. General housekeeping guidelines.

- ✓ Work areas to be kept clear of combustible material.
- ✓ Do not obstruct fire equipment, exits and aisles.
- ✓ Do not store material under staircases.
- ✓ Clean up spills regularly.
- ✓ Store flammable liquids in approved flammable liquid stores.

## STUDY UNIT 4

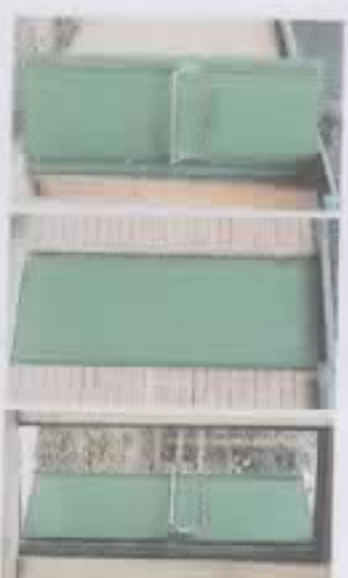
### FIRE FIGHTING EQUIPMENT

#### LEARNING OUTCOMES

On completion of this study unit the student will....



- Know the different types of fire extinguishers.
- Know the use of the different types of fire extinguishers.
- Identify the correct fire extinguisher to fight a specific class of fire.
- How to operate a fire extinguisher.
- Know the location and operation of a fire hose.



• Dry Powder Extinguisher. (Multi-Purpose)



Blue

**Best for:**

Fires involving organic solid materials such as wood, cloth, paper, plastics, Coal etc.

**Danger:**

Safe on live electrical equipment, although does not penetrate the spaces in equipment easily and the fire may re-ignite. This type of extinguisher does not cool the fire very well and care should be taken that the fire does not flare up again. Smouldering material in deep seated fires such as upholstery or bedding can cause the fire to start up again. Do not use on domestic chip or fat pan fires.

**How to use:**

Point the jet or discharge horn at the base of the flames and, with a rapid sweeping motion, drive the fire towards the far edge until all the flames are out. If the extinguisher has a shut-off control wait until the air clears and if you can still see the flames, attack the fire again

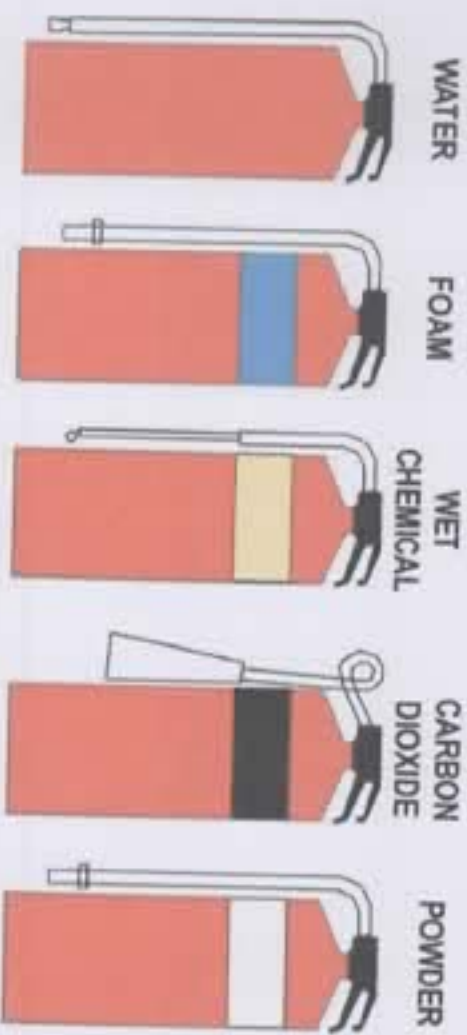
**How it works:**

Similarly to almost all extinguishing agents the powders acts as a thermal ballast making the flames too cool for the chemical reactions to continue. Some powders also provide a minor chemical inhibition, although this effect is relatively weak. These powders thus provide rapid knockdown of flame fronts, but may not keep the fire suppressed.



1. Know the use of different types of fire extinguishers

There are five different types of fire extinguishers they are colour coded and the instruction labels indicate on which type of fires we can use them. The pictogram indicate the use if the pictogram is crossed out it is not suitable for that class of fire



• WATER EXTINGUISHER



Signal Red

**Best for:**

Fires involving organic solid materials such as wood, cloth, paper, plastics, Coal etc.

**Danger:**

Do not use on burning fat or oil or on electrical appliances.

**How to use:**

Point the jet at the base of the flames and keep it moving across the area of the fire. Ensure that all areas of the fire are out.

**How it works:**

Water has a great effect on cooling the fuel surfaces and thereby reducing the pyrolysis rate of the fuel.



- Carbon Di-Oxide Extinguisher

### BLACK

#### Best for:

Live electrical equipment when it is not possible to isolate the electric supply and flammable liquids such as grease, fats, oil paint, petrol etc. but not on domestic chip or fat pan fires.

#### Danger:

Do not use on domestic chip or fat pan fires. This type of extinguisher does not cool the fire very well and you need to watch that the fire does not start up again. Fumes from CO2 extinguishers can be harmful if used in confined spaces: ventilate the area as soon as the fire has been controlled.

#### How to use:

The discharge horn should be directed at the base of the flames and the jet kept moving across the area of the fire.

#### How it works:

Carbon dioxide extinguisher works on classes "B" and "C" and works by suffocating the fire. Carbon dioxide will not burn and displaces air.



- Foam Extinguisher. (AFFF)

### Cream (Aqueous film-forming foam) (multi-purpose)

#### Best for:

Fires involving solids. Liquids such as grease, fats, oil, paint, petrol, etc but not on domestic chip or fat pan fires. Can be used on class A fires but not recommended.

#### Danger:

Do not use on domestic chip or fat pan fires.

#### How to use:

For fires involving solids, point the jet at the base of the flames and keep it moving across the area of the fire. Ensure that all areas of the fire are out. For fire involving liquids, do not aim the jet straight into the liquid. Where the liquid on fire is in a container, point the jet at the inside edge of the container or on a nearby surface above the burning liquid. Allow the foam to build up and flow across the liquid.

#### How it works:

They are mainly water based, with a foaming agent so that the foam can float on top of the burning liquid and break the interaction between the flames and the fuel surface.



2. How to operate a fire extinguisher?



It's easy to remember how to use a fire extinguisher if you remember the acronym **PASS**

- Pull
- Aim
- Squeeze
- Sweep

Fires can be very dangerous and you should always be certain that you will not endanger yourself or others when attempting to put out a fire.

For this reason, when a fire is discovered...

**REACT** upon discovery of fire or smoke

- ♦ Remove persons in immediate danger!
- ♦ Ensure doors are closed! (*confine fire/smoke*)
- ♦ Activate the building alarm!
- ♦ Call the Fire Department!
- ♦ Treat ALL fires as DANGEROUS!



- Wet chemical



**Canary Yellow**

**Best for:**

The specialist wet chemical extinguishers are ideal for Class "F" fires, involving cooking oils and fats, such as lard, olive oil, sunflower oil, maize oil and butter.

**Danger:**

Check manufacturer's instructions for suitability of use. These extinguishers are not recommended for class B fires and home use.

**How to use:**

Apply the wet chemical using the extended applicator in slow circular movements, which give a gentle, yet highly effective application. Apply the fine spray onto the burning fat until the surface of the burning cooking oil changes into a soapy like substance, this then prevents re-ignition. The gentle application helps prevent hot oil splashing onto the user.

**How it works:**

Most class "F" extinguishers contain a solution of potassium acetate, sometimes with some potassium citrate or potassium bicarbonate. The extinguishers spray the agent out as a fine mist. The mist acts to cool the flame front, while the potassium salts saponify the surface of the burning cooking oil, producing a layer of foam over the surface. This solution thus provides a similar blanketing effect to a foam extinguisher, but with a greater cooling effect. The saponification only works on animal fats and vegetable oils, so class F extinguishers cannot be used for class B fires. The misting also helps to prevent splashing the blazing oil. Tests have established that a 6 litre extinguisher is capable of extinguishing a fire in a 75 litre capacity deep fat fryer. The extinguisher is easy to use producing a gentle but highly effective spray.

## STUDY UNIT 5

### FIGHTING FIRE

#### LEARNING OUTCOMES

On completion of this study unit the student will....



- Make the correct decision in the process of fighting a fire.
- Stop a fire by using the correct equipment and procedure.
- Apply general precautions when fighting a fire.
- Identify and apply the correct procedures after a fire has been put out accordingly to SOP.

### 3. Fire hoses

- The large Fire hose reels located in multi storey buildings are to be used by building occupants to fight fire, especially if they are trapped and cannot escape to an emergency exit.
- The Fire Hose Reels are connected to the mains water supply and extend for about 30m. Some Fire Hose Reels are located in cabinets while others are visible on a wall in a hall or corridor. There will always be appropriate signage, indicating their location.



#### How to operate a Fire Hose

Different Fire Hose Reels are very similar in operation, here is the generic procedure:

- Ensure the nozzle or jet is in the closed position.
- Turn on the main valve (some will not let the nozzle out until this is done).
- Pull the hose off the drum towards the fire.
- Open the nozzle or valve and direct the stream towards the fire.

### 3. Fighting the fire

- Maintain a safe distance. Remember the effective extinguisher range.
- Ensure you know what is burning and what type of extinguisher to use.
- Switch off electricity.
- Know how to use the extinguisher.
- Never turn your back on the fire.
- Move around the perimeter of the fire to maximize coverage of the extinguisher agent.
- Overhaul ensures that every piece of burning material is completely extinguished. To prevent rekindling of the fire:
- Locate hidden burning material. Extinguish and safely remove it.
- Remove heat by cooling.
- If the fire gets out of control leave the area immediately.
- Always fight the fire with the exit to your back.



### 1. Make the correct decision in the process of fighting a fire

**HOWEVER - BEFORE DECIDING TO FIGHT THE FIRE, KEEP THESE THINGS IN MIND:**

- Know what is burning. If you don't know what's burning, you won't know what kind of extinguisher to use.
- Even if you have an ABC fire extinguisher, there may be something in the fire that is going to explode or produce toxic fumes.

### 2. Fighting a fire

When a fire starts you only have a few seconds to re-act in those seconds you must respond quickly and effectively.

Do not fight the fire if:

- You have called the fire department first
- There is a possibility that your path of escape may be blocked
- The fire is spreading fast
- You do not know what is burning
- You don't know how to use the equipment
- You don't have adequate or appropriate equipment. If you don't have the correct type or large enough extinguisher, it is best not to try fighting the fire.
- You might inhale toxic smoke. When synthetic materials such as the nylon in carpeting or foam padding in a sofa burn, they can produce hydrogen cyanide, acrolein, and ammonia in addition to carbon monoxide. These gases can be fatal in very small amounts.
- Do not attempt to suppress a fire that is clearly too large for the equipment at hand
- Your instincts tell you not to. If you are uncomfortable with the situation for any reason, just let the fire department do their job.



6. ROAST

In case of fire, remember to ROAST:

**R**eport

Call 911 to report the fire.

**O**bserve

Locate the location of the fire.

**A**bort

If possible, shut off available sources of fuel or oxygen.

**S**pray

If necessary, spray sparks with water or fire extinguishers.

**T**ake off

Have employees exit the building in a safe way.

7. Methods of extinguishment

- ♦ Cooling
- ♦ Starvation
- ♦ Smother

Extinguishment of Temperature Reduction cooling

One of the most common methods of extinguishment is by cooling with water. The process of extinguishment by cooling is dependent on cooling the fuel to a point where it does not produce sufficient vapor to burn. If we look at fuel types and vapor production, we find that solid fuels and liquid fuels with high flash points can be extinguished by cooling. Low flashpoint liquids and flammable gases cannot be extinguished by cooling with water as vapor production cannot be sufficiently reduced. Reduction of temperature is dependent on the application of an adequate flow in proper form to establish a negative heat balance.



4. Procedure after the fire

- ♦ Watch the fire area for re-ignition.
- ♦ If the fire flares up again report.
- ♦ Call the fire department to inspect fire area.
- ♦ Replace used equipment.

5. Handing over to the emergency services

Inform the officer taking over the following:

- ♦ First indication where the fire started.
- ♦ Development of the fire before arrival.
- ♦ Any actions taken before arrival.
- ♦ Action of last person on site.
- ♦ Colour of smoke.
- ♦ Wind direction.
- ♦ Time the fire started.
- ♦ Nearby hazards such as LPG known to you that can affect operations.

